

The emergence of the unmarked in L2 acquisition: Interpreting null subjects

Tomoko Monou

`monou@otsu.icl.keio.ac.jp`

Shigeto Kawahara

`kawahara@rci.rutgers.edu`

Tokyo Conference on Psycholinguistics 2013

Keio University

March 9th, 2013

Background

- The acquisition of L2 is significantly affected by L1 knowledge; a.k.a. L1 transfer [18].
- However, there are observations that L1 transfer does not entirely govern L2 acquisition, hinting at the role of grammar in L2 acquisition (see [20] for a review).

Background

- This talk focuses on the emergence of the unmarked (TETU) effect in L2 acquisition.
- It has been observed in the phonological literature that L2 learners go through a stage in which they only show unmarked structures, albeit the lack of evidence in L1 or L2 [1, 4, 5].

Question

- Would we observe the effect of the emergence of the unmarked in L2 acquisition in domains other than phonology? (cf. [2, 5, 8].)
- In order to address this question, the current experiment tested the case of the interpretations of null subjects where L1 transfer and TETU make different predictions.
- The experiment demonstrates that at least the majority of speakers show behaviors that are compatible with the markedness theory.

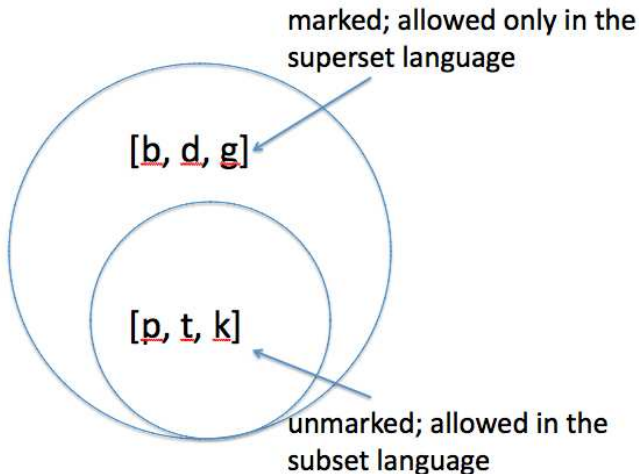
What is markedness?

- “Markedness” [13] has been used to express several different notions, not all of which are well received or well defined [10].
- We use the notion of markedness which can be defined formally.

What is markedness?

- Implicational universal: The linguistic structure X is marked and Y is unmarked, if all languages that allow Y also allow X, but not vice versa [8, 9, 17].
- In terms of the subset-superset relationship: The unmarked structure X is an element in the subset languages (also allowed in the superset languages) [3, 7, 19] cf. [15].

Illustration of markedness

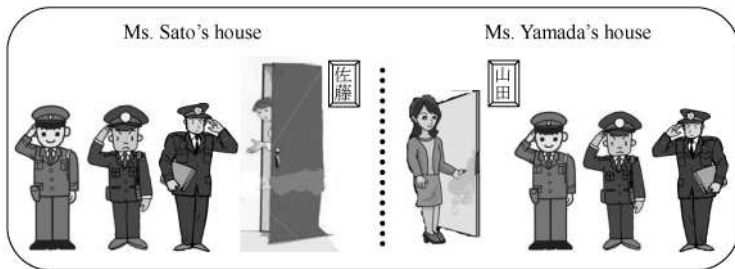


A case study: Japanese null subject construction

- (1) a. San-nin-no keisatu-ga Sato-san-no
three-CL-no police-officer-NOM Ms.Sato-GEN
ie-ni kita.
house-DAT came
'Three police officers came to Ms.Sato's house.'
- b. e Yamada-san-no ie-ni-mo kita.
NULL Ms.Yamada-GEN house-DAT-also came
'The three police officers/Three police officers also
came to Ms. Yamada's house.'

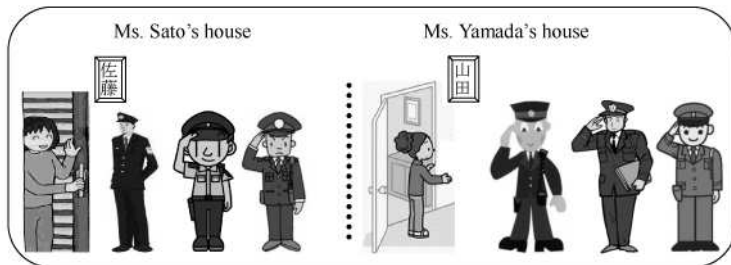
This sentence is ambiguous in terms of the interpretation of null subjects [16].

Pro reading



The three policemen who came to Ms. Yamada's house are the same as those who came to Ms. Sato's house.

Quantificational reading



The three policemen who came to Ms. Yamada's house can be different from those who came to Ms. Sato's house.

The Mandarin Chinese equivalent

- (2) a. Sǎn-ge jǐngchá lái-le Zuǒtáng
three-CL police-officer come-ASP Ms.Sato's
jiǎ,
house
'Three police officers came to Ms.Sato's house.'
- b. e yě lái-le Shāntián jiǎ.
NULL also come-ASP Ms. Yamada's house
'The three police officers also came to
Ms.Yamada's house.'

This sentence in Mandarin Chinese is not ambiguous: it only allows the pro reading.

The pro reading is unmarked

- Japanese allows both of these two readings, while Mandarin Chinese allows only the pro reading.
- A cross-linguistic survey with 9 languages with null subject reveals [14] :
 - 1 languages with both readings,¹
 - 2 languages with only the pro reading,²
 - 3 but no languages with only the quantificational reading.

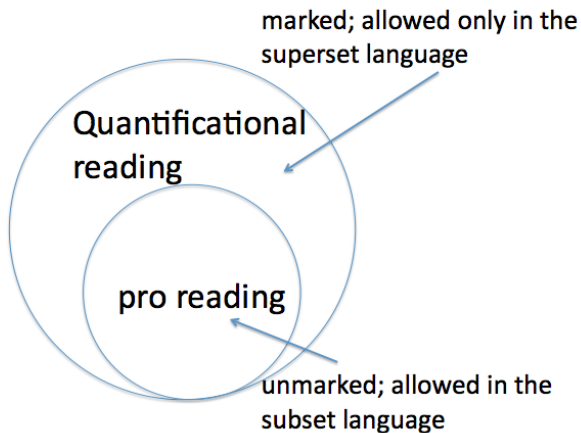
¹Korean, Japanese, and Greek.

²Basque, Mandarin Chinese, Serbo-Croatian, Spanish, and Turkish.

The pro reading is unmarked

- Accordingly, the pro reading is the **unmarked** reading (or the “subset reading”), while the quantificational reading is the **marked** reading (or the “superset reading”).
- (This markedness asymmetry may be grounded in an additional processing burden of introducing new referents for the quantificational reading (cf. [8, 11, 12]), but we nevertheless take the position that this psycholinguistic difficulty is grammaticalized via some abstraction [14].)

Illustration of markedness



The predictions of the two theories

- When Japanese speakers (the superset speakers) learn Mandarin Chinese (the subset languages) as L2:
- L1 transfer: Japanese speakers would accept both readings in Mandarin Chinese sentences, since their L1 (=Japanese) allows both readings.
- TETU (or the Subset Principle): Japanese speakers start with the unmarked, subset reading (i.e. the pro reading).

Participants

- The participants were 22 introductory-level Japanese learners of MC.
- The experiment targeted introductory learners to tap the initial L2 learning state.

Participants

- The age of first exposure to MC ranged from 18;7 to 19;5 (average: 18;11) (well after the critical period).
- The duration of exposure of formal instruction in Japan ranged from 0;10 to 1;10. (average: 0;11, about a year).

Stimuli

- Target stimuli: five sets of Mandarin Chinese sentences with null subjects.
- Control stimuli: five sets of equivalent Japanese sentences.
- Those who did not understand the ambiguity in the Japanese control sentences were excluded from the analysis.

Task

- A truth value judgment task (a well-established approach in the acquisition literature [7]; see [14] for detail).
- A picture that depicts a particular interpretation (examples shown previously) was shown along with each target sentence.

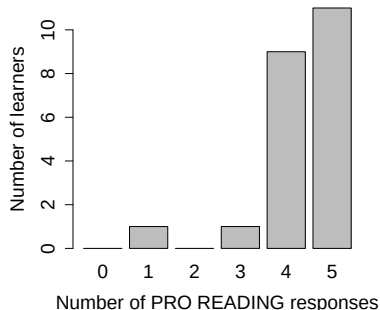
Task

- The participants were asked to indicate whether each sentence correctly described the picture.
- The two questions on the same sentence were presented separately (i.e. one question per trial). The task was thus not to detect an ambiguity of stimulus sentences.

Procedure

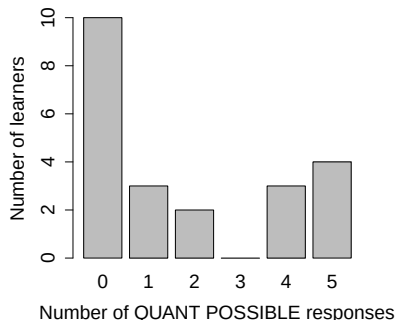
- For each sentence, the picture that indicates a quantificational interpretation was presented before the picture that indicates a pro reading.
- It was expected to be easier for native MC speakers and Japanese learners to assign a pro interpretation than a quantificational reading, and a pro reading may prime the quantificational reading for the same sentence.

Results: Pro reading



Almost all the participants judged the pro reading to be possible.

Results: Quant-reading

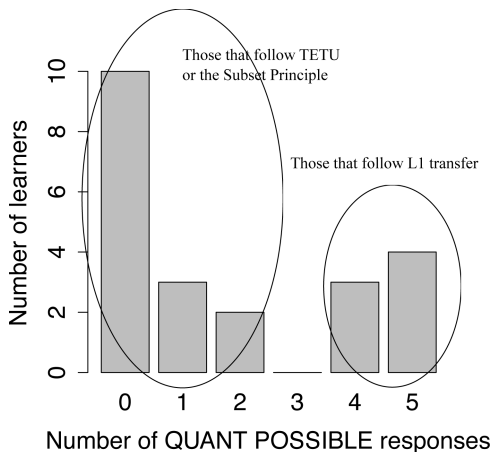


Many learners rejected the quantificational reading; the difference between the two readings is significant ($p < .001$ by a Wilcoxon test).

Discussion

- The TETU effect (or the Subset Principle) predicts this behavior; i.e., those who accept only the unmarked, pro reading.
- However, some other learners accepted both readings (the marked and unmarked).
- L1 transfer may have governed the behavior of these speakers.

Discussion: two groups of participants?



Discussion: a question

- Could they have learned the lack of quantificational reading in Mandarin Chinese?
 1. Their textbook does not state that the pro-reading is how to interpret null subjects in Mandarin Chinese.
 2. Neither do their language instructors teach that.
 3. The participants all reported that they had never been explicitly how to interpret null subjects in Mandarin Chinese.

Further questions

- Could it have been that the participants simply did not like ambiguity?
- Unlikely, because they did not have a trouble detecting the ambiguity for Japanese sentences (i.e. the control stimuli).

Further questions

- Could it have been that the quantificational reading imposed more processing burden, and hence it was rejected?
- No evidence that this purported difference in processing burden influences their judgement when they judged L1 sentences.³

³After all, we do not know yet that the quantificational reading imposes substantially more psycholinguistic burden to process than the pro reading in this context. This assumption needs to be shown in future experimental work (cf. [6]).

More remaining questions and issues

- Why do the current participants split into two groups?
- Not clear. At least their age and duration of exposure to L2 are comparable across all participants, which therefore cannot be the separating factor.

More remaining questions and issues

- Small number of items (five in the current experiment)?
- Order effect? The current experiment was blocked for a reason, but it could have introduced a compound.

Summary

- L1 transfer and some linguistic mechanisms (markedness pressure or the Subset Principle) can conflict.
- In the case of the interpretation of null subjects, many speakers showed behaviors that are compatible with the grammatical principle(s) (the emergence of the unmarked or the Subset Principle)—their behavior cannot be explained by L1 transfer, at least.
- Some other speakers behaved as predicted by L1 transfer.

Acknowledgements

Thanks to Makiko Hirakawa, Julien Musolino, Yukio Otsu, William O'Grady, and members of TPL for valuable comments and suggestions. We attribute the remaining errors to the evil spirit.

References

- [1] Altenberg, Evelyn & Robert Vago (1983) Theoretical implications of an error analysis of second language phonology production. *Language and Learning* **33**: 427–447.
- [2] Ayoun, Dalila (1996) The subset principle in second language acquisition. *Applied Psycholinguistics* **17**(2): 185–213.
- [3] Berwick, Robert (1985) *The acquisition of syntactic knowledge*. Cambridge, MA: MIT Press.
- [4] Broselow, Ellen, Su-I. Chen, & Chilin Wang (1998) The emergence of the unmarked in second language phonology. *Studies in Second Language Acquisition* **20**: 261–280.
- [5] Broselow, Ellen & Daniel Finer (1991) Parameter setting in second language phonology and syntax. *Second Language Research* **7**: 35–59.
- [6] Crain, Stephen & Mark Steedman (1985) On not being led up the garden-path: the use of context by the psychological parser. In *Natural Language Processing*, David Dowty, Lauri Karttunen, & Arnold Zwicky, eds., Cambridge: Cambridge University press.

- [7] Crain, Stephen & Rosalind Thornton (1998) *Investigations into Universal Grammar: A guide to experiments on the acquisition of syntax and semantics*. MIT Press.
- [8] Eckman, Fred (2004) Universals, innateness and explanation in second language acquisition. *Studies in Language* **28**: 682–703.
- [9] Greenberg, Joseph (1978) Some generalizations concerning initial and final consonant clusters. In *Universals of Human Language*, vol. 2: Phonology, Joseph Greenberg, ed., Stanford: Stanford University Press, 243–280.
- [10] Haspelmath, Martin (2006) Against markedness (and what to replace it with). *Journal of Linguistics* **42**: 25–70.
- [11] Hawkins, John & Anne Cutler (1988) Psycholinguistic factors in morphological asymmetry. In *Explaining Language Universals.*, J. A. Hawkins, ed., Oxford: Basil Blackwell, 280–317.
- [12] Hawkins, John A. (1992) Innateness and function in language universals. In *The evolution of human languages*, John A. Hawkins & Murray Gell-Mann, eds., Reading, MA: Addison-Wesley Publishing Company, 87–120.

- [13] Jakobson, Roman (1941) *Kindersprache, Aphasie, und allgemeine Lautgesetze*. Uppsala: Almqvist & Wiksell [English translation: Child Language, Aphasia and Phonological Universals, Mouton], translated into English by A. Keiler, 1968.
- [14] Monou, Tomoko (2013) *Restrictive second language development paths: Evidence from subject ellipsis constructions in Japanese and Mandarin Chinese*. Doctoral dissertation, Keio University.
- [15] Musolino, Julien (2006) On the semantics of the Subset Principle. *Language learning and Development* 2(3): 195–218.
- [16] Oku, Satoshi (1998) *A theory of selection and restriction in the Minimalist perspective*. Doctoral dissertation, University of Connecticut.
- [17] Prince, Alan & Paul Smolensky (1993/2004) *Optimality Theory: Constraint Interaction in Generative Grammar*. Malden and Oxford [originally circulated in 1993 as ms. University of Colorado and Rutgers University]: Blackwell.
- [18] Schwartz, Bonnie D. & Rex Sprouse (1996) L2 cognitive states and the Full Transfer/Full Access model. *Second Language Research* 12: 40–72.

- [19] Wexler, Kenneth (1993) The subset principle is an intensional principle. In *Knowledge and Language (vol 1), From Orwell's problem to Plato's problem*, Eric Reuland & Abraham W., eds., Netherlands: Kluwer Academic Publishers, 217–239.
- [20] White, Lydia (2003) *Second Language Acquisition and Universal Grammar*. Cambridge: Cambridge University Press.